

**Table S8. Nonlinear civil-photoperiod classifications**

|                                               | Qualifying transition | Qualifying transition bracket (h) | Endpoint derivative              | Interpretation                                                 |
|-----------------------------------------------|-----------------------|-----------------------------------|----------------------------------|----------------------------------------------------------------|
| <b>Light level</b>                            |                       |                                   |                                  |                                                                |
| Mean melEDI                                   | Yes                   | 16.10–16.20                       | 0.008 (95% CI -0.111 to 0.127)   | Detected increase followed by a sustained zero-compatible tail |
| Brightest 10 h mean                           | Yes                   | 15.17–15.27                       | 0.059 (95% CI -0.083 to 0.201)   | Detected increase followed by a sustained zero-compatible tail |
| Darkest 10 h mean                             | Yes                   | 14.86–14.96                       | -0.086 (95% CI -0.216 to 0.044)  | Detected increase followed by a sustained zero-compatible tail |
| <b>Duration and continuous period</b>         |                       |                                   |                                  |                                                                |
| Time above 1,000 lx melEDI                    | Yes                   | 14.86–14.96                       | 0.098 (95% CI -0.176 to 0.371)   | Detected increase followed by a sustained zero-compatible tail |
| Time above 250 lx melEDI during wake          | Yes                   | 14.35–14.45                       | 0.060 (95% CI -0.159 to 0.279)   | Detected increase followed by a sustained zero-compatible tail |
| Time below 10 lx melEDI before sleep          | No                    | —                                 | -0.085 (95% CI -0.154 to -0.016) | No preceding detected increase                                 |
| Time below 1 lx melEDI during sleep           | No                    | —                                 | 0.062 (95% CI -0.042 to 0.165)   | No preceding detected increase                                 |
| Longest continuous period above 250 lx melEDI | No                    | —                                 | 0.041 (95% CI 0.025 to 0.058)    | Increase remained detected at the recorded maximum             |
| <b>Exposure history</b>                       |                       |                                   |                                  |                                                                |
| melEDI dose                                   | Yes                   | 14.24–14.35                       | 0.070 (95% CI -0.069 to 0.209)   | Detected increase followed by a sustained zero-compatible tail |

A 'Yes' is the formal derivative-defined classification: the immediately preceding point has an interval wholly above zero, the next contains zero, and every later interval through the recorded maximum remains zero-compatible. Endpoint derivatives and 95% confidence intervals are on each model's linear-predictor scale, in model-scale units per hour of civil photoperiod.

**Supplementary Table S8.** Derivative-based classifications for nine near-eye personal light-exposure metrics across the observed civil-photoperiod range. A qualifying transition required the preceding fitted-slope interval to be wholly above zero, the next interval to include zero and every later interval through the recorded maximum to remain zero-compatible. These classifications are descriptive and do not identify a physiological or environmental ceiling.